

# CONTROL SYSTEMS LABORATORY

|                         |               |
|-------------------------|---------------|
| <b>Experiment No:</b>   | 3             |
| <b>Experiment Name:</b> | Experiment 3  |
| <b>Student No:</b>      | 151220192027  |
| <b>Student Name:</b>    | Mustafa Yağcı |
| <b>Date:</b>            | 10.05.2023    |

## 1. OBJECTIVES

- 1-Simplify the Block Diagrams to obtain a closed-loop transfer function of the dynamics.
- 2-Obtain a Xcos model for a given dynamical system.
- 3-Solve a given differential equation by using Octave.

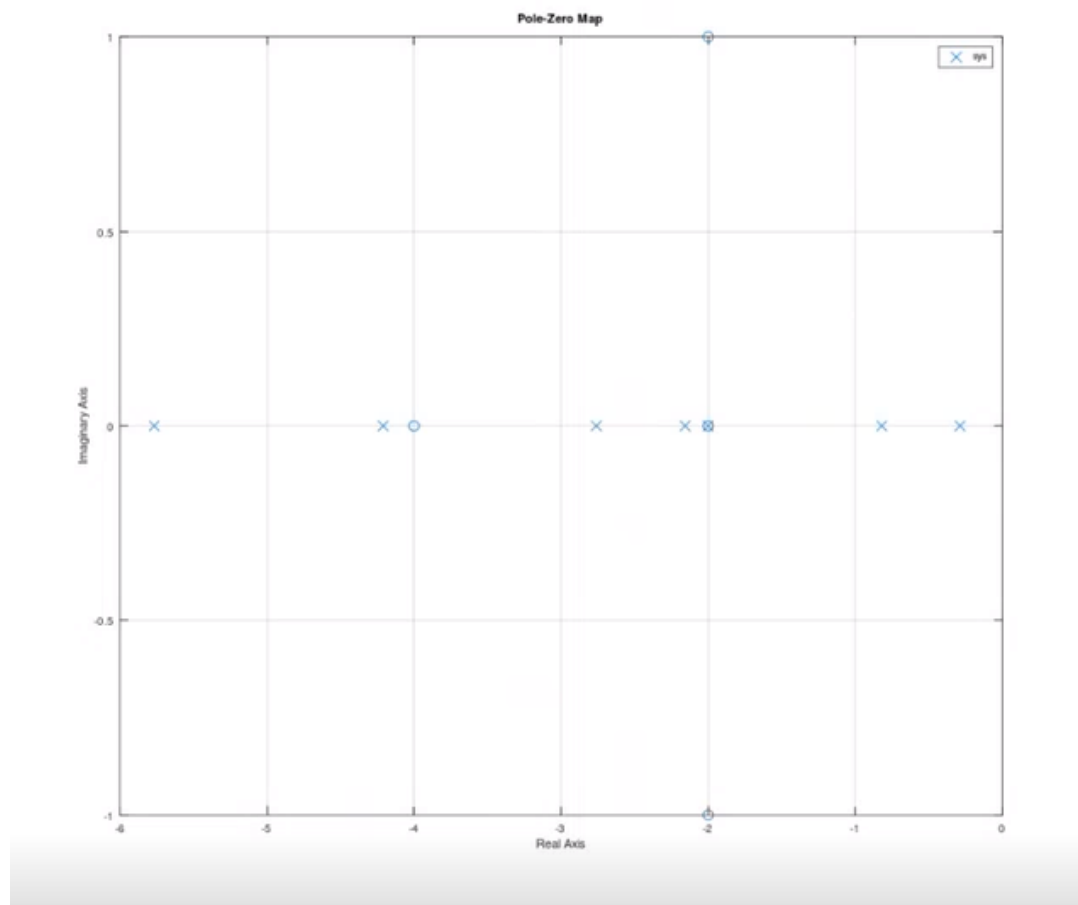
## 2. EXPERIMENTAL RESULTS

- 1-We have found the transfer function from the block diagram.

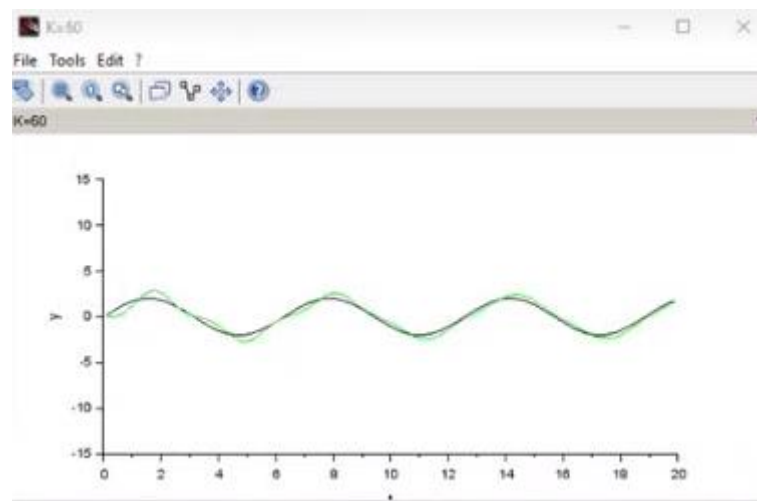
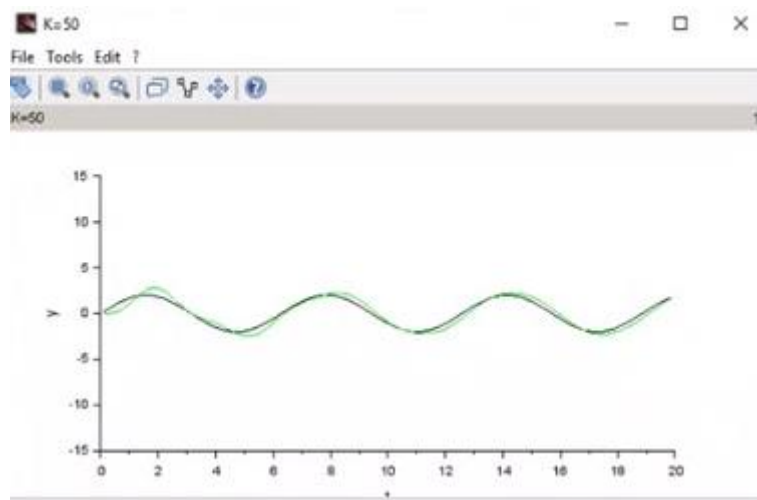
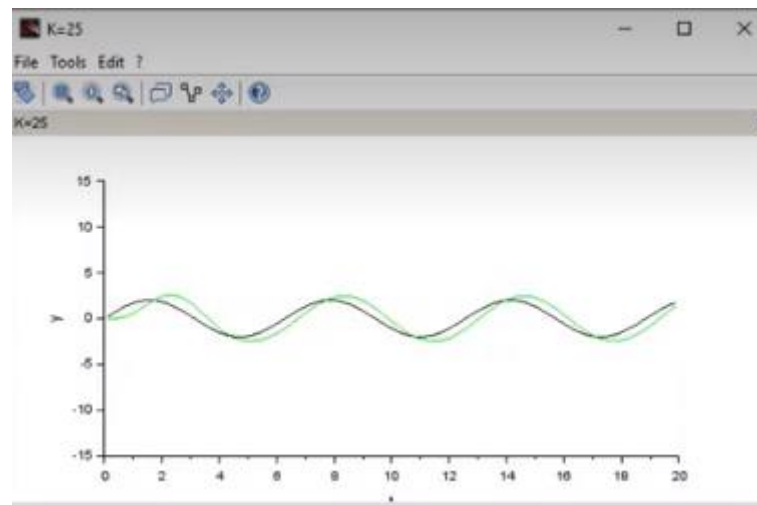
```
Transfer function 'sys' from input 'u1' to output ...  
yl: 
$$\frac{2 s^4 + 20 s^3 + 74 s^2 + 124 s + 80}{s^7 + 18 s^6 + 128 s^5 + 462 s^4 + 901 s^3 + 924 s^2 + 438 s + 68}$$

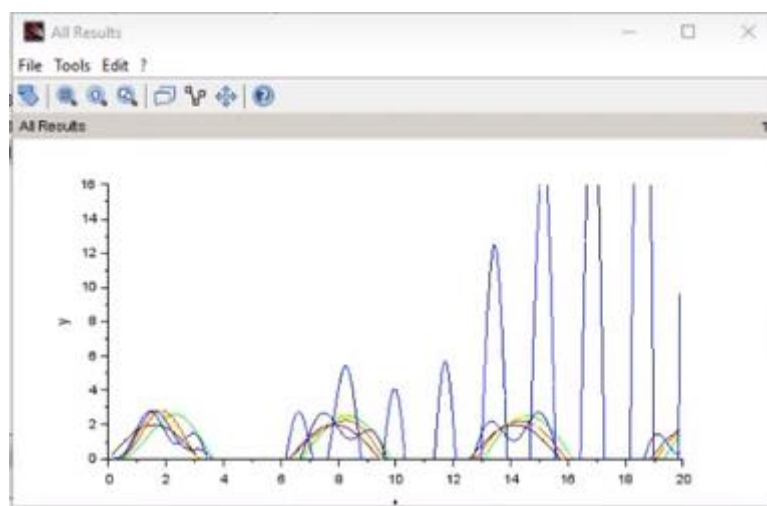
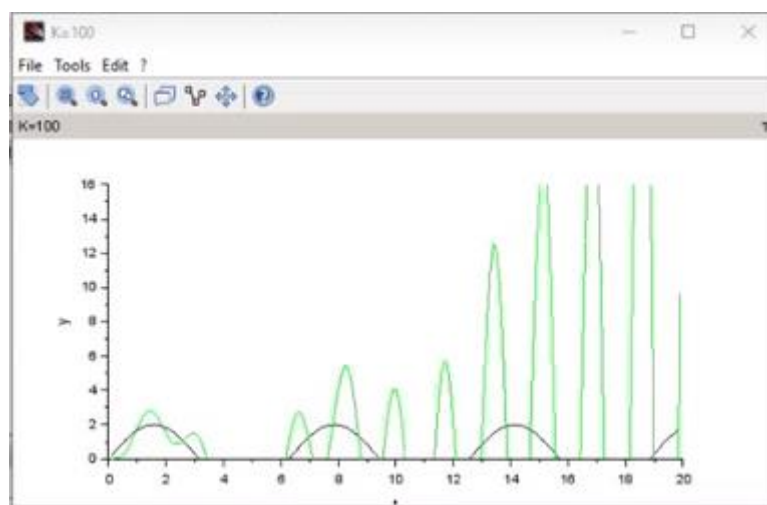
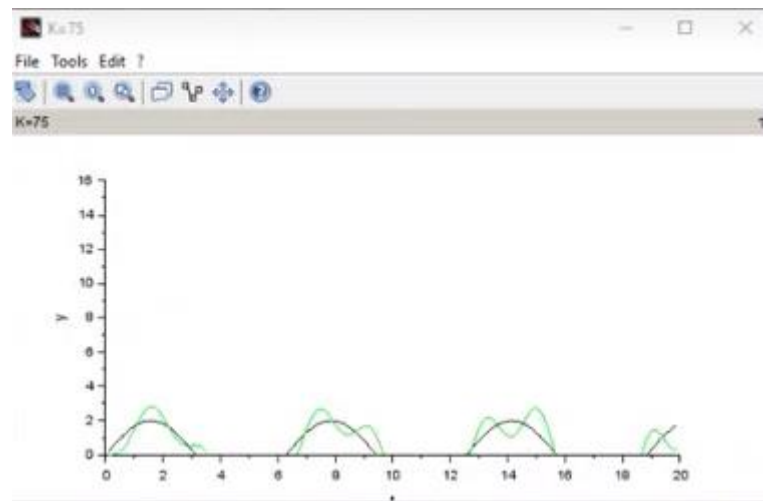
```

- 2-We have found the pole-zero map of the transfer function.



3- Values of k.





4-Stability analysis.

| RH Table |                  |                                      |
|----------|------------------|--------------------------------------|
| $s^3$    | 1                | 10                                   |
| $s^2$    | 7                | K                                    |
| $s^1$    | $\frac{70-K}{7}$ | $\frac{7 \times 10 - 1 \times K}{7}$ |
| $s^0$    | K                |                                      |

$$\frac{70-K}{7} > 0 \quad \rightarrow \quad 70 - K > 0 \quad \rightarrow \quad K - 70 < 0 \quad \rightarrow \quad K < 70$$

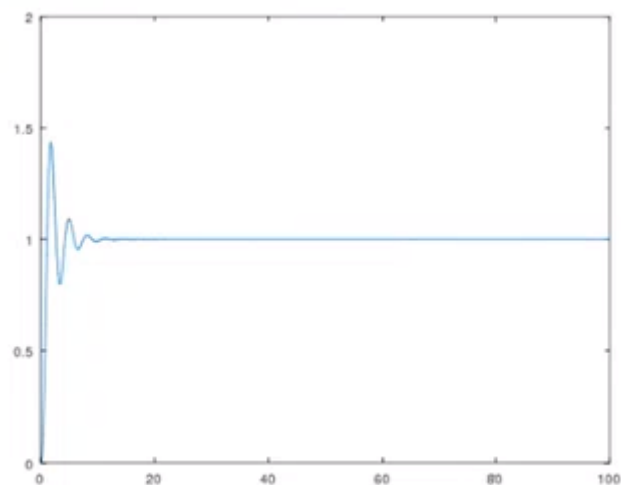
|                   |              |
|-------------------|--------------|
| Stable            | $0 < K < 70$ |
| Unstable          | $K > 70$     |
| Marginally Stable | $K = 70$     |

$$K = 70 \quad \rightarrow \quad 7s^2 + K = 0 \quad \rightarrow \quad 7s^2 = -70 \quad \rightarrow \quad s^2 = -10 \quad \rightarrow \quad s = \pm\sqrt{-10} = \pm 3.16228j$$

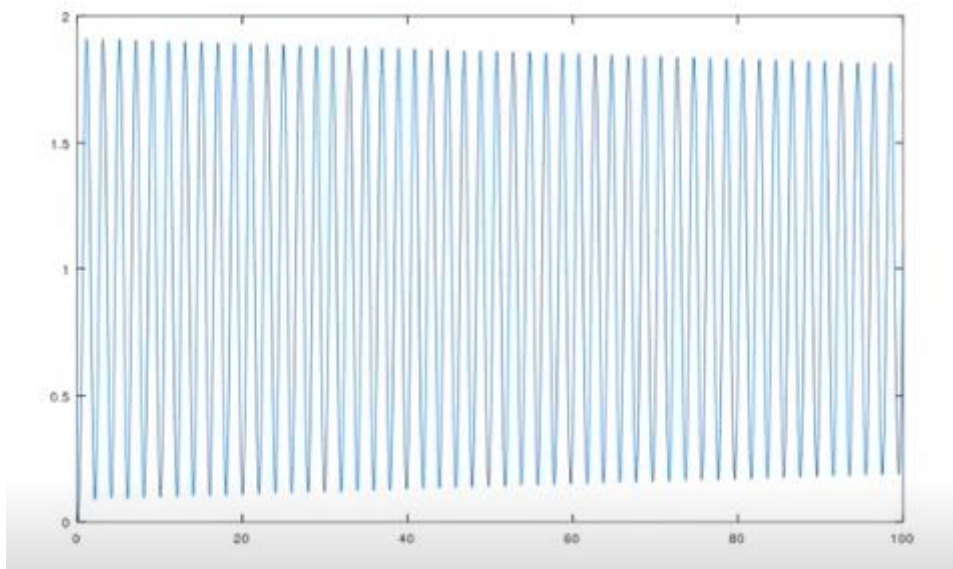
$$w_n = 3.16228$$

5-

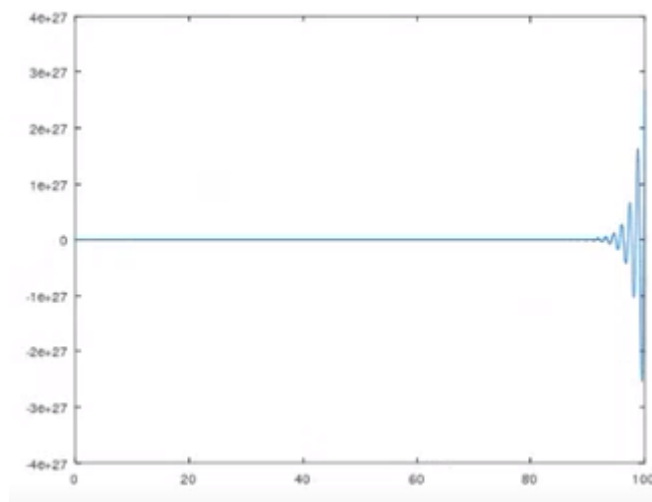
k=25



$K=70$



$K=170$



### 3. CONCLUSION & DISCUSSION

As a result, If the value of  $k$  is in the range up to 70, the system is stable, but beyond that point, the system becomes unstable.

## **REFERENCES**

Write your reference list in this section by using numbering with square brackets - []. You should follow IEEE Citation StyleGuide for referencing format.

- [1] IEEE Citation Style Guide, Available: <http://www.ijssst.info/info/IEEE-Citation-StyleGuide.pdf>, Access Date: October 2021